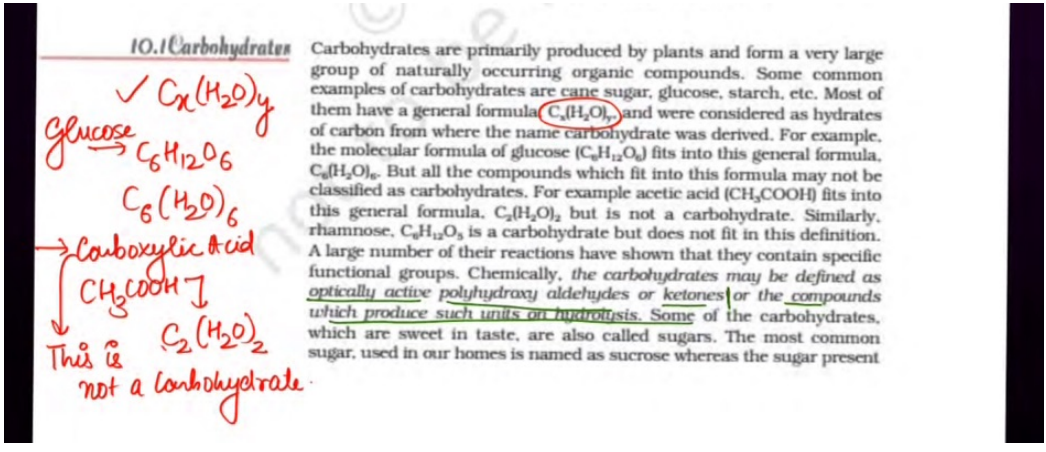
12th hacker series

Biomolecules

introduction

BIOMOLECULES

- Carbohydrate
  - glucose
  - fructose
  - disaccharands
  - starch
- Protein
  - Amino Acid
- Nucleic acid
- Hormones
  Vitamins
  enzyms



10.1 Carbohydrates

Carbohydrates are primarily produced by plants and form a very large group of naturally occurring organic compounds. Some common examples of carbohydrates are cane sugar, glucose, starch, etc. Most of them have a general formula $C_x(H_2O)_y$ and were considered as hydrates of carbon from where the name carbohydrate was derived. For example, the molecular formula of glucose ($C_6H_{12}O_6$) fits into this general formula, $C_6(H_2O)_6$. But all the compounds which fit into this formula may not be classified as carbohydrates. For example acetic acid (CH_3COOH) fits into this general formula, $C_2(H_2O)_2$ but is not a carbohydrate. Similarly, rhamnose, $C_6H_{12}O_5$ is a carbohydrate but does not fit in this definition. A large number of their reactions have shown that they contain specific functional groups. Chemically, *the carbohydrates may be defined as optically active polyhydroxy aldehydes or ketones or the compounds which produce such units on hydrolysis.* Some of the carbohydrates, which are sweet in taste, are also called sugars. The most common sugar, used in our homes is named as sucrose whereas the sugar present

✓ $C_x(H_2O)_y$

Glucose → $C_6H_{12}O_6$

$C_6(H_2O)_6$

→ Carboxylic Acid

CH_3COOH

$C_2(H_2O)_2$

This is not a Carbohydrate.

Carbohydrates

- the carbohydrates may be defined as optically, active, polyhydroxy aldehyde or ketones or the compound which produce such units on hydrolysis
- Some of the carbohydrates, which are sweet in taste are also called sugars

eg – Glucose $C_6H_{12}O_6$ → polyhydroxy aldehyde

Fructose $C_6H_{12}O_6$ → polyhydroxy ketone

eg - Sucrose $\xrightarrow{\text{hydrolysis}}$ glucose + Fructose

Classification of carbohydrates

Monosaccharides

- a carbohydrate that cannot be hydrolysed further to give simpler unit of polyhydroxy aldehyde or ketone is called a monosaccharide.
- Some common examples are glucose, fructose, ribose, etc

Oligosaccharides

- carbohydrates hat yield 2 to 10 monosaccharide units on hydrolysis are called oligosaccharides
- They are further classified as disaccharides, trisaccharides, tetrasaccharide , etc. Depending upon the number of Monosaccharide , they provide on hydrolysis.
- Amongst these, the most common are disaccharides
- The two monosaccharide units obtained on hydrolysis of the disaccharides may be same or different
- For example, one molecule of sucrose on hydrolysis gives one molecule of glucose and one molecule of fructose, whereas maltose gives two molecules of only glucose

Polysaccharides

- carbohydrates, which yield a large number of monosaccharide units on hydrolysis are called polysaccharides
- some common examples are starch, cellulose, glycogen, gums, etc
- Polly saccharides are not sweet in taste, hence they are also called non-sugars

The carbohydrates can also be classified as either reducing or non-reducing

- all those carbohydrates, which reduce fehling's solution, and tollen's reagent are referred to as reducing sugar
- All monosaccharides, whether Aldose or ketose are reducing sugar

Classification of carbohydrates

Monosaccharides

- monosaccharides are further classified on the basis of number of carbon atoms and the functional group present in them
- if a monosaccharide contains an aldehyde group, it is known as aldose and if it contains a keto group, it is known as ketose

Carbon atoms	General term	Aldehyde	Ketone
3	Triose	Aldotriose	Ketotriose
4	Tetrose	Aldotetrose	Ketotetrose
5	Pentose	Aldopentose	Ketopentose
6	Hexose	Aldohexose	Ketohexose
7	Heptose	Aldoheptose	Ketoheptose

Glucose

- glucose occurs freely in nature as well as in the combined form
- it is present in sweet fruits and honey
- Ripe grapes also contains glucose in large amounts
- Mechanism of preparation of glucose:
- 1) from sucrose (cane sugar)

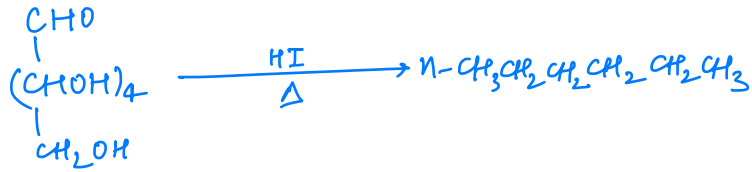
$$\underset{\text{sucrose}}{C_{12}H_{22}O_{11}} \xrightarrow[\text{in alcoholic soln}]{\text{dil HCl or } H_2SO_4} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

- 2) from starch- commercially glucose is obtained by hydrolysis of starch by boiling it with dilute H2SO4 at 393 Kelvin under pressure

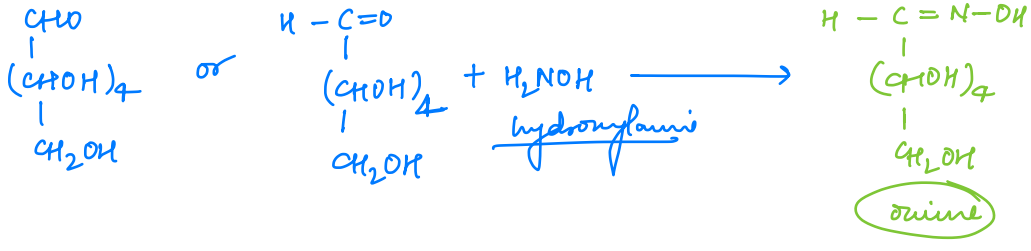
$$(C_6H_{10}O_5)_n \xrightarrow{\text{hydrolysis}} n\,C_6H_{12}O_6$$

- glucose is an Aldohexose and is also known as dextrose
- it is the monomer of many of the larger carbohydrates, namely starch, cellulose
- it is probably the most abundant organic compound on the earth
- it was assigned the structure given below on the basis of the following evidences
- 1) its molecular formula was found to be C6H1206

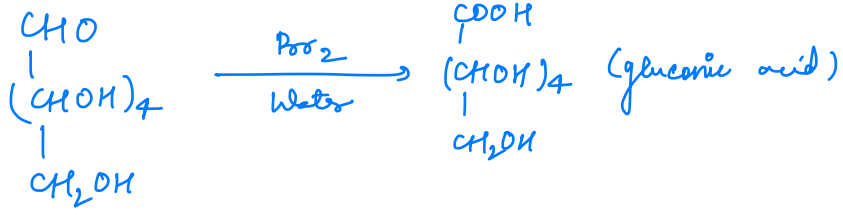
- 2) on prolonged heating with HI, it forms n- hexane, suggesting that all the six carbon atoms are linked in a straight chain

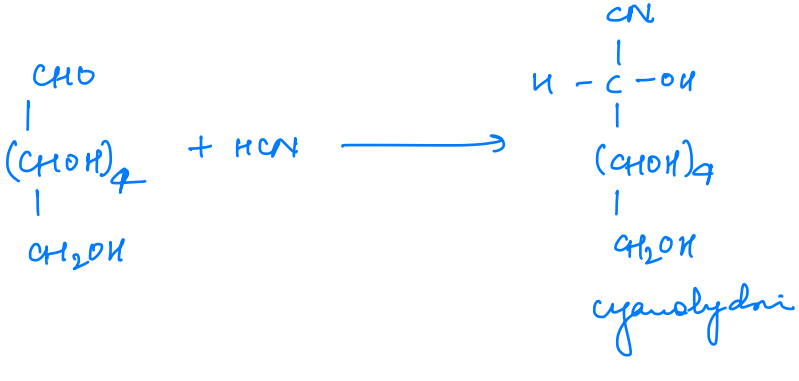


- glucose reacts with hydroxylamine to form an oxime and adds a molecule of hydrogen cyanide to give cyanohydrin . This reaction. confirm the presence of carbonyl group in glucose.

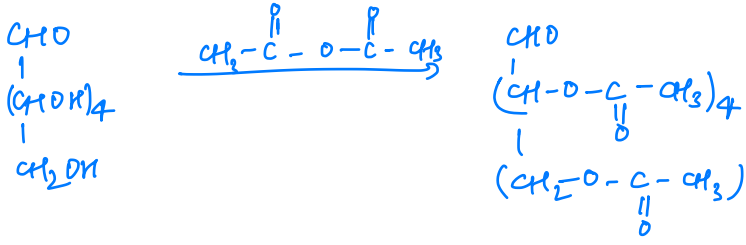


- glucose gets oxidised to 6 carbon carboxylic acid (gluconic acid) on reaction with a mild oxidising agent like bromine water
- This indicates that the carbonyl group is present as an aldehydic group

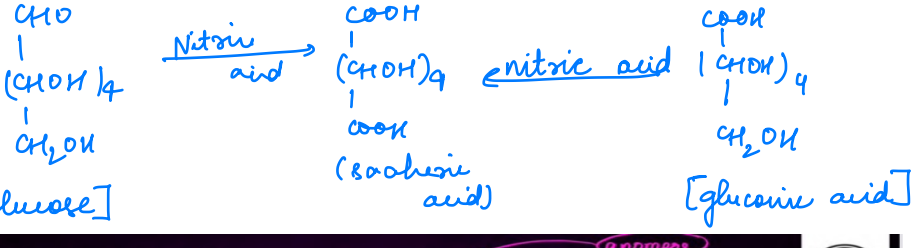


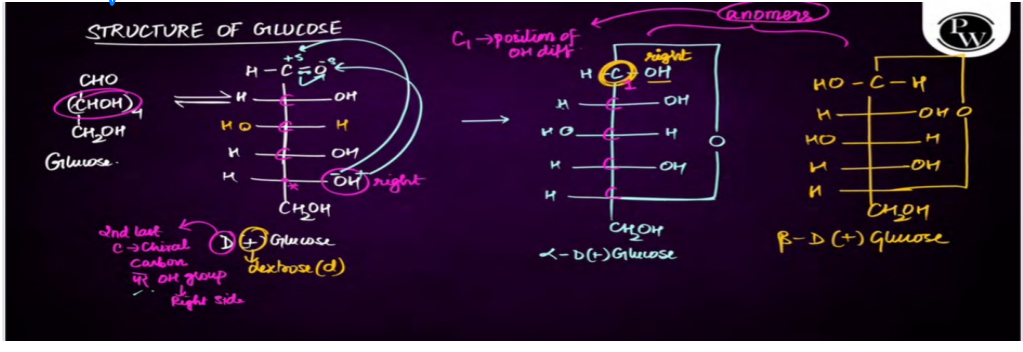


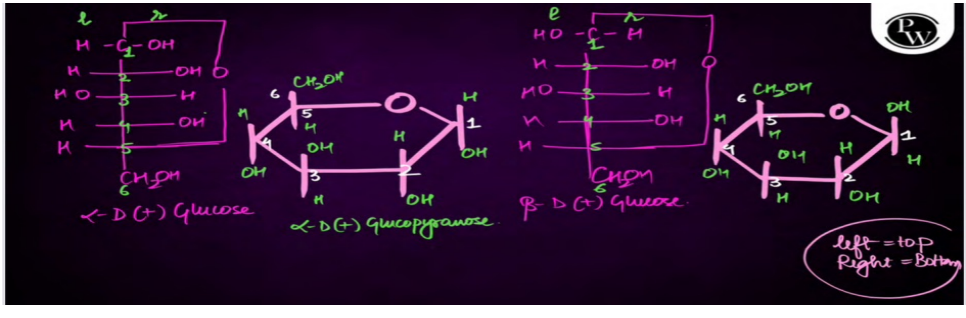
- acetylation of glucose with acidic anhydride gives glucose pantaacetate, which confirms the presence of 5 OH groups
- Since it exist as a stable compound. 5 OH groups should be attached to different carbon atoms.



- oxidation with nitric acid , glucose as well as gluconic acid, both yield or dicarboxylic acid, saccharic acid.
- this indicates the presence of primary alcoholic group in glucose

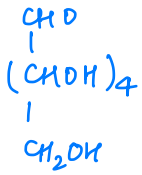






Open chain structure

- molecular formula of glucose is experimentally found as $C_6H_{12}O_6$



- epimers: they are pair of diastereomers that differ only in the configuration about a single carbon
- anomers: they are diasteromers that differ in the configuration at the acetal or hemiacetal C atom of the sugar in its cyclic form or anomers are epimers, whose confirmation differ only about C1
- for example- α-D-(+) & β-D(+) glucose are anomers, α-D(−) and β-D-(−)-fructose are anomers

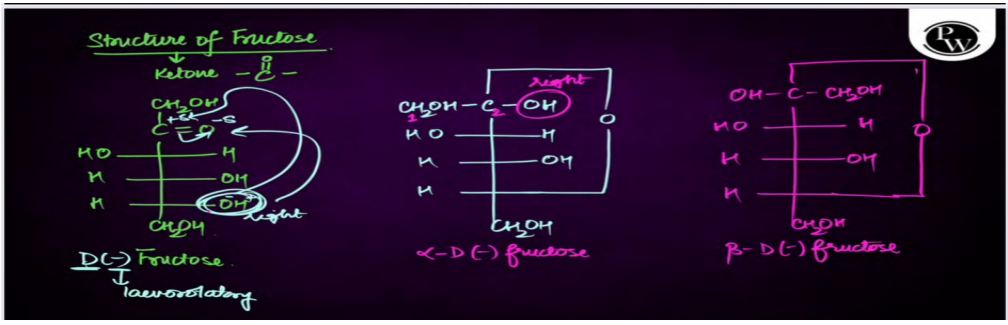
Fructose

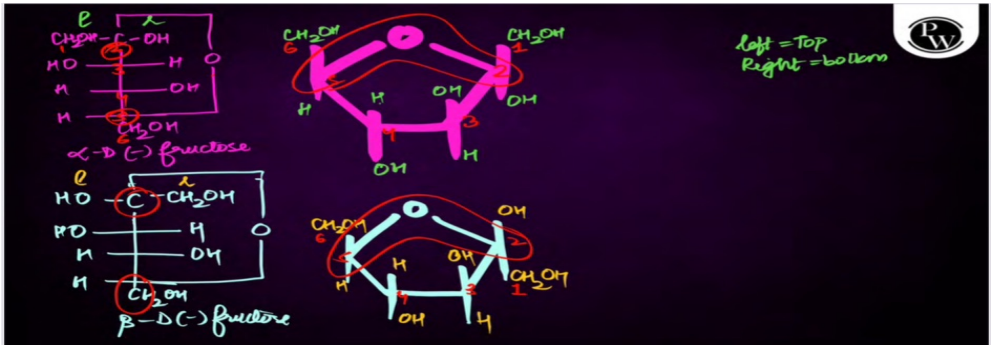
- It occurs both in combined as well as free state
- Fructose is named as fruit sugar because it is present in honey and more sweet fruits in free state
- it is the sweetest monosaccharide and present in cane sugar and insulin and combined state
- it is also known as alpha- Laevulose i.e, naturally occurring fructose is laevorotatory
- Preparation
- 1) by acid hydrolysis of cane sugar

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow[\text{Warm}]{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$$

- 2) by enzymatic action on sucrose

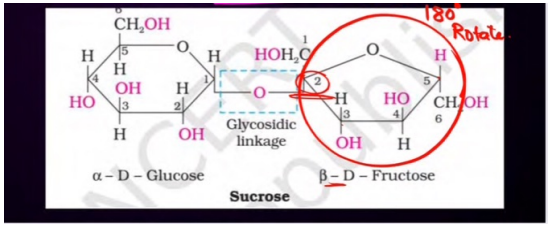
$$C_{12}H_{22}O_{11} \xrightarrow{\text{invertase}} \text{glucose} + \text{fructose}$$

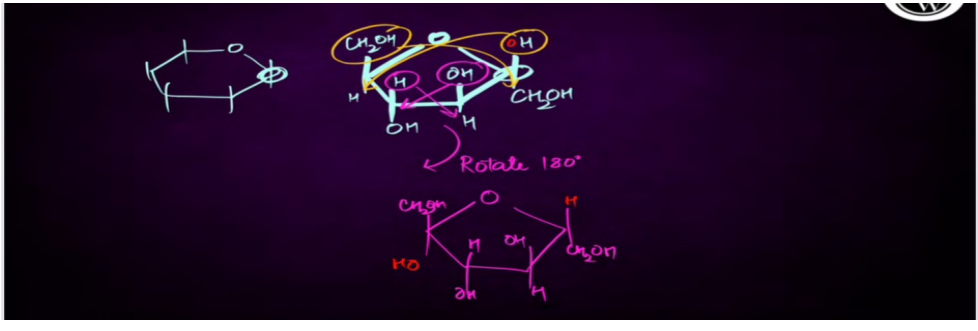


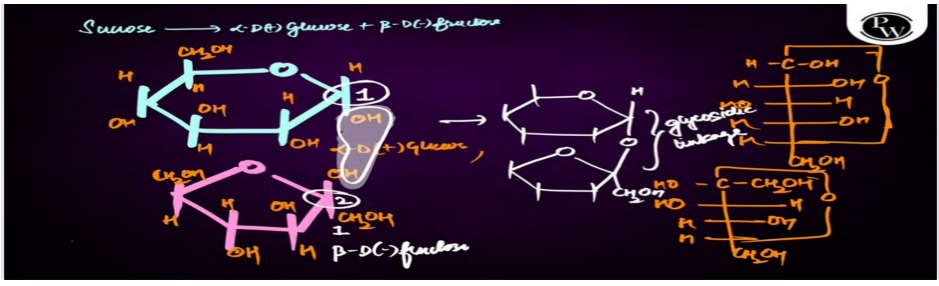


Sucrose

- these two monosaccharides are held together by a glycosidic linkage between C1 of α-D glucose and C2 of β-d fructose
- Since the reducing group of glucose and fructose are involved in glucosidic bond formation, sucrose is a non-reducing sugar

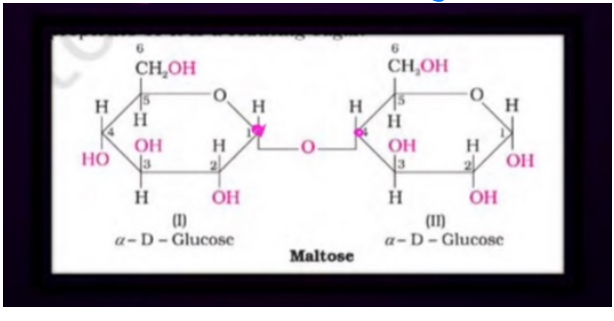






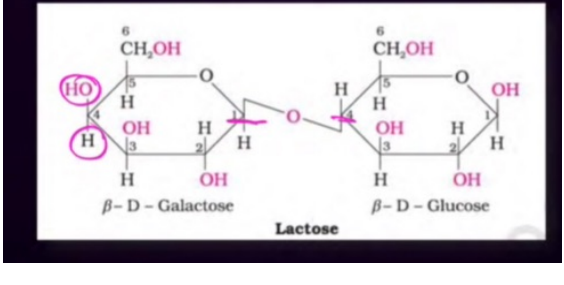
Maltose

- another disaccharide maltose is composed of 2 D-glucose units in which C1 of one glucose
- 1) is linked to C4 of another glucose unit
- 2) The free aldehyde group can be produced at C1 of the second glucose in solution and it shows reducing property, so it is a reducing sugar



Lactose

- it is more commonly known as milk sugar since this disacchride is found in milk
- it is composed of beta D glucose and beta D galactose
- The linkage is between C1 of galactose and C4 of glucose
- Free aldehyde group may be produced at C1 of glucose unit, hence it is also reducing sugar





Polysaccharides

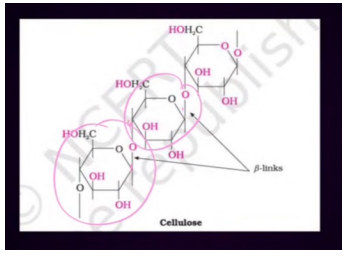
Starch

- it is the main storage polysaccharide of plants
- it is the most important dietary source for human beings
- High content of starch is found in cereals, roots, tubes, and some vegetables
- it is a polymer of d-glucose and consist of two components, amylose and amylopectin

- amylose is water is soluble component which constituents about 15 to 20% of starch
- Chemically, amylose is a long branch chain with 200 to 1000 d-glucose units held together by C1 - C4 glycosidic linkage

Cellulose

- cellular occurs exclusively in plants, and it is the most abundant organic substance in plant kingdom
- it is a predominant constituent of cell wall in plant cells
- it is a straight chain polysaccharide composed of only beta D glucose units which are joined by glycosidic linkage between the C1 of one glucose unit and C4 of the next glucose unit



Glycogen

- the carbohydrates are stored in animal body as glycogen
- it is also known as animal starch because it structure is similar to amylopectin and is rather more highly branched
- it is present in the liver, muscles and brain
- When the body needs, glucose enzymes, break the glycogen down to glucose
- Glycogen is also found in yeast and fungi

Protein

- they occur in every part of the body and form the fundamental basis of structure and function of life
- they are also required for growth and maintenance of the body
- The word protein is derived from the greek " proteios " which means primary or of prime importance
- All proteins are polymers of amino acid



Classification of amino acid

- depending on the relative number of amino (- NH2) and carboxyl (- COOH) functional group, amino acid are classified as acidic, basic and neutral
- More number of amino group, make it basic and more carboxyl group make it acidic , equal number of amino and carboxyl group, make it neutral

Properties of amino acid

- except glycine, all other naturally occurring amino acids are optically active since Alpha carbon is asymmetric
- L amine acids are represented by writing the $-NH_2$ Group on the left hand side.

- in actual sense acidic properties due to COOH group and basic properties due to NH2 group because it exist as zwitter ion

Zwitter ions

- due to presence of an acidic and basic group in the same molecule, most of the amino acids are neutral and exist largely as dipolar or zwitter ions or inner salts in which proton from the carbon group has been transferred to the amino group and thus a dipolar ion containing both a positive and a negative charge is formed

Peptide linkage

- amino acid units are linked together by peptide linkage and form polypeptides
- in a polypeptide structure, free amino group i.e, N terminal residue is written on the left hand side and the free carboxyl group on the right hand side of the chain

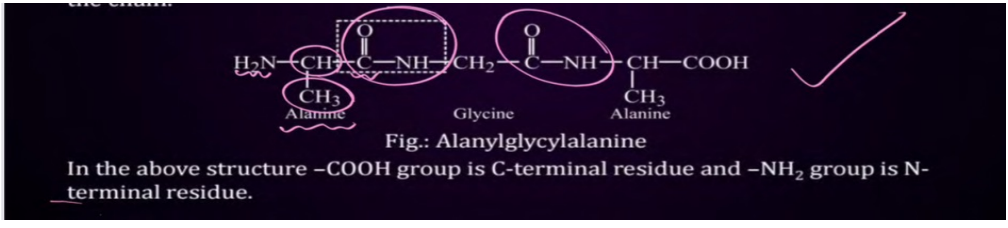


Fig.: Alanylglycylalanine

In the above structure –COOH group is C-terminal residue and $-NH_2$ group is N-terminal residue.

Essential and non-essential amino acid

- the amino acids which can be synthesised in the body are known as non-essential amino acids
- on the other hand, those which cannot be synthesised in the body and must be obtained through diet are known as essential amino acid

Classification of proteins

On the basis of molecular structure

- 1) fibrous protein- when polypeptide chain run parallel and are held together by hydrogen and disulphide bonds, then fibre like structures are formed. Such proteins are generally insoluble in water . example- keratin, collagen, myosin, etc

- 2) globular protein- the poly peptide chain in these proteins is folded around itself to form spheroidal shape. The folding takes place in a manner such that - hydrophobic parts are pushed inwards and hydrophilic parts are pushed outward
- as a result, water molecules interact strongly with the polar groups and hence globular proteins are soluble in water. As compare to the fibrous proteins, these are very sensitive to small changes in temperature and pH . example- insulin, albumin, antibodies, haemoglobin, fibrinogens, etc

On the basis of chemical composition (not in NCERT)

- 1) simple proteins: these proteins on hydrolysis give only amino acids. example - albumin in the white portion of eggs, keratin in hair, nails, horns, etc
- 2) conjugated proteins- these are the proteins in which the protein part is combined with non-protein part. These proteins on hydrolysis give a non-protein in addition to the amino acids. The main function of the prosthetic group is to control the biological function of the protein. Prosthetic groups are carbohydrate, phosphate group, lipids, and so on. Caesin of milk, haemoglobin of blood are conjugated proteins
- 3) derived proteins: these are the degradation products obtained by partial hydrolysis of simple or conjugated proteins with acids, alkali or enzymes. Example- proteases, peptones, and polypeptide

Proteins

- primary structure of protein: proteins may have one or more polypeptide chains
- Each polypeptide is a protein has amino acid linked with each other in a specific sequence, and it is the sequence of amino acids that is said to be the primary structure of that proteins. Any change in this primary structure I.e, the sequence of Amino acid It creates a different protein.

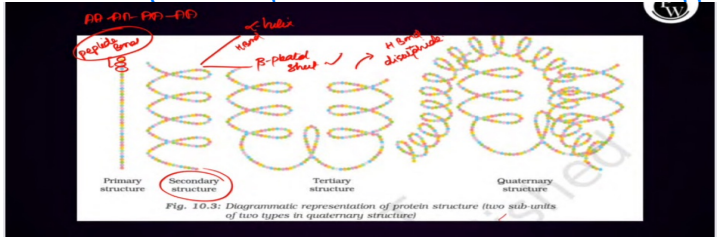
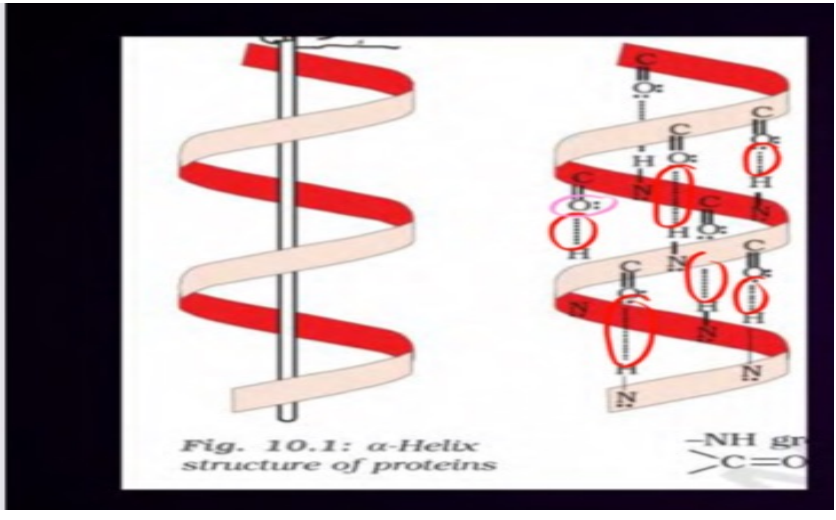


Fig. 10.3: Diagrammatic representation of protein structure (two sub-units of two types in quaternary structure)



- the secondary structure of protein: refers to the shape in which long polypeptide chain can exist
- They are found to exist in two different types of structures- alpha helix, and beta plated sheets structures
- These structures arises due to regular folding of the backbone of the polypeptide chain due to hydrogen bonding between the NH group of the peptide bond
- Helix is one of the most common ways in which a poly
- peptide chain forms all possible hydrogen bonds by twisting into a right hand screw (helix) with the NH group of amino acid residue hydrogen bond to the CO of an adjacent turn of the helix

- tertiary structure of proteins: the tertiary structure of proteins represents overall folding of the polypeptide chain i.e, further folding of the secondary structure
- it gives rise to 2 major molecular shapes viz fibrous, and globular
- The main forces which stabilise the 2° and 3°, structure of proteins are hydrogen, bond, disulphide, linkages, Vanderwaals , and electrostatic forces of attraction

- quarterary structure of proteins: some of the proteins are composed of two or more polypeptide chain referred to as subunits
- The special arrangement of these sub units with respect to each other is known as quaternary structure

Denaturation of proteins

- a protein found in a biological system. Having a unique three dimensional structure and a specific biological activity is called a native protein.
- When a protein in its native form is subjected to a physical change like change, temperature or pH, the hydrogen bonds are disturbed
- as a result, globules unfold and Helix get uncoiled and protein lose is biological activity
- This is called denaturation of proteins
- During denaturation, secondary and tertiary structures are destroyed, but primary structures remain intact
- Eg- coagulation of egg white, curdling of milk

Enzymes

- enzymes are very specific for a particular reaction and for a particular substrate
- They are generally named after the compound or class of compounds upon which they work
- For example- the enzyme that catalyse hydrolysis of maltose into glucose is named as Maltase

Composition of enzymes (not in NCERT)

- chemically, all enzymes are globular proteins
- However, some enzymes are also associated with some non-protein component called the co-factor for their activity
- These co-factors are of two types
- 1) inorganic ions – Zn^{2+}, Mg^{2+}, Mn^{2+}
- 2) organic molecules- these are also of two types
- I) coenzymes: these are small organic molecules which are loosely held to the protein and can be easily separated by dialysis.

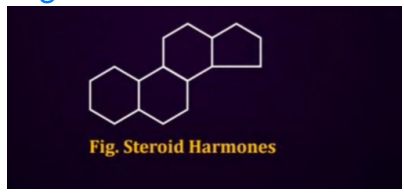
- II) prosthetic group: these are also organic molecules which are tightly held to protein by covalent bonds, but can be separated by only careful hydrolysis. Most of these are derived from vitamins such as biotin. Enzymes are vital for biological process without the life process would be very slow.

Hormones

- these chemicals, which are secreted by ductless glands and transported to different parts of the body by the blood stream, where they control different physiological actions of the body
- The word hormone is derived from greek language, which means to gear up or to excite
- Hormone control cell and tissue growth, heartbeat, blood pressure, secretion of digestive enzymes, kidney function, reproductive system, and lactation, etc
- in mammals, the secretion of the hormones is controlled by anterior lobe of the pituitary gland located at the base of the brain
- These hormones are then carried to other gland, such as adrenal cortex and sex glands to stimulate the production of other hormones

Classification of hormones

- on the basis of structure and composition, hormones are classified into the following types
- 1) steroid hormones- these hormones are composed of steroid nucleus having the following structure


Fig. Steroid Harmones

- these are mostly secreted by testes and adrenal context of males, and hence called as adrenal Cotex hormones or sex hormones
- Example of these hormones are androgen , testosterone, and dihydrotestisterone
- These hormones are stimulant of male sex characteristic
- in females, sex hormones are oestrogens. They are produced by ovaries.

- 2) polypeptide hormones- these hormones are polypeptide
- These hormones are secreted by the posterior lobe of the pituitary gland. Example oxytocin , vasopressin
- 3) amino hormones: these are water soluble hormones containing amino groups. These are derived from amino acids, example thyroxine and adrenaline.

Nucleic acid

- the particle in nucleus of the cell responsible for heredity are called chromosomes, which are made up of proteins, and another type of bio molecules called nucleic acids
- These are mainly of two types, DNA and RNA
- Since nucleic acids are long chain polymers of nucleotides, so there are also called polynucleotide
- DNA contains four bases viz adenine, guanine, cytosine, and Thymine
- RNA also contains four basis. The first three basis are same as in DNA, but the fourth one is uracil

Structure of nucleic acid

- a unit formed by the attachment of a base to one position of sugar is known as nucleoside
- in nucleosides, the sugar carbon are numbered as 1, 2, 3 etc. In order to distinguish these from bases
- When nucleoside is linked to phosphoric acid at five position of sugar moiety , we get a nucleotide

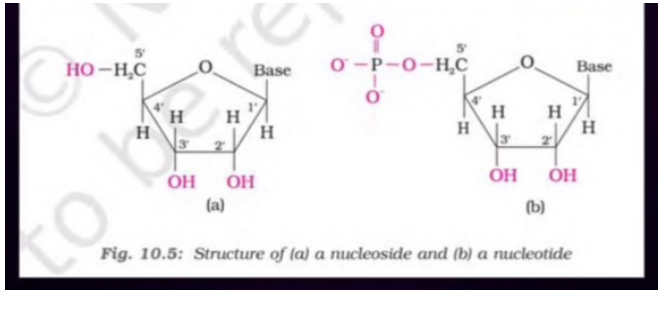


Fig. 10.5: Structure of (a) a nucleoside and (b) a nucleotide

QUESTION

Q1

The general formula of carbohydrates is

$C_x(H_2O)_y$ Ex. $C_6H_{12}O_6$ → $C_6(H_2O)_6$ → $C_n(H_2O)_n$

A. $C_nH_{2n+1}O$

B. $C_nH_{2n}O$

C. $C_n(H_2O)_n$ ✓

D. $C_n(H_2O)_{2n}$

QUESTION

Q2

Match the sugars in column I with their types given in column II and mark the appropriate choice.

	Column I		Column II
(A)	Glucose → aldohexose	(i)	Ketohexose
(B)	Fructose	(ii)	Aldohexose
(C)	Ribose	(iii)	Aldotetrose
(D)	Erythrose	(iv)	Aldopentose

A. (A) → (iv), (B) → (i), (C) → (iii), (D) → (ii)

B. (A) → (iii), (B) → (iv), (C) → (i), (D) → (ii)

C. (A) → (i), (B) → (ii), (C) → (iii), (D) → (iv)

D. (A) → (ii), (B) → (i), (C) → (iv), (D) → (iii)

QUESTION

Q3

On oxidation with a mild oxidising agent like Br_2/H_2O, the glucose is oxidised to

A. saccharic acid

B. glucaric acid

C. gluconic acid ✓

D. valeric acid

QUESTION

Q4

The letter 'D' in carbohydrates signifies

A. dextrorotatory (d) ✗

B. configuration ✓

C. diamagnetic nature

D. mode of synthesis

QUESTION

Q5

The anomeric carbon in D(+) glucose is

A. C-1 carbon ✓

B. C-2 carbon

C. C-5 carbon

D. C-6 carbon

QUESTION

The α- and β- forms of glucose are

Q6.

A isomers of D(+) glucose and L(–) glucose respectively

B diastereomers of glucose

C anomers of glucose

D isomers which differ in the configuration of C-2.

QUESTION

How many C-atoms are there in a pyranose ring?

A 3

B 5

C 6

D 7

QUESTION

Which of the following is a non-reducing sugar?

A Glucose

B Sucrose

C Maltose

D Lactose

QUESTION

Which one of the following sets of monosaccharides forms sucrose?

A α-D-galactopyranose and α-D-glucopyranose

B α-D-glucopyranose and β-D-fructofuranose

C β-D-glucopyranose and α-D-fructofuranose

D α-D-glucopyranose and β-D-fructopyranose.

QUESTION

What are the hydrolysis products of sucrose?

A Fructose + Fructose

B Glucose + Glucose

C Glucose + Galactose

D Glucose + Fructose

QUESTION

The glycosidic linkage involved in linking the glucose units in amylose part of starch is

A $C_1 - C_4$ β-linkage

B $C_1 - C_6$ β-linkage

C $C_1 - C_6$ α-linkage

D $C_1 - C_4$ α-linkage

QUESTION

What is the basic formula of starch?

A $(C_6H_{12}O_6)_n$

B $(C_6H_{10}O_5)_n$

C $C_{12}H_{22}O_{11}$

D $(C_6H_{12}O_4)_n$

QUESTION

Starch is composed of two polysaccharides which are

A amylopectin and glycogen

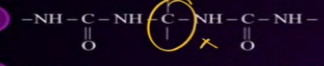
B amylose and glycogen

C amylose and amylopectin

D cellulose and glycogen

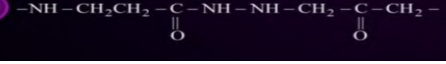
QUESTION

Which of the following represents a peptide chain?

A 

B 

C 

D 

QUESTION

Which of the following is the correct statement?

A Starch is a polymer of α-glucose

B Amylose is not a component of starch.

C Proteins are composed of only one type of amino acid

D In cyclic structure of fructose, there are five carbon and one oxygen atoms.

QUESTION

In fibrous proteins, polypeptide chains are held together by

A. van der Waals forces
B. electrostatic forces of attraction
C. hydrogen bonds
D. covalent bonds

QUESTION

Secondary structure of protein refers to

A. sequence of amino acids in polypeptide chain
B. Bonds between alternate polypeptide chains
C. Folding patterns of polypeptide chain
D. Bonding between NH_3^+ and COO^- of two peptides.

QUESTION

Most common types of secondary structures of proteins are

A. α-helix and β-helix structures
B. α-helix and β-pleated sheet structures
C. right and left hand twisted structures
D. globular and fibrous structures.

QUESTION

Which is a fat soluble vitamin?

KEDA

A. Vitamin A
B. Vitamin B_6
C. Vitamin C
D. Vitamin B_2

QUESTION

Bases common to RNA and DNA are

A. adenine, guanine, cytosine
B. adenine, uracil, cytosine
C. adenine, guanine, thymine
D. guanine, uracil, thymine

QUESTION

In DNA, the complimentary bases are

(A) uracil and adenine, cytosine and guanine
(B) adenine and thymine, guanine and cytosine
(C) adenine and thymine, guanine and uracil
(D) adenine and guanine, thymine and cytosine

QUESTION

A unit in nucleic acid which contains 'base-sugar phosphate' unit is called

(A) nucleotide
(B) nucleoside
(C) phosphotide
(D) polypeptide

QUESTION

Which of the two components of starch is water soluble? [All India 2020, 2014]

Amylose

QUESTION

What is the basic structural difference between glucose and fructose? [All India 2019C]

RCHO RCOR

QUESTION

Write the products obtained after hydrolysis of lactose. [Delhi 2019]

OR

What are the products of hydrolysis of lactose? [All India 2013; Delhi 2010C]

galactose + glucose

QUESTION

Write the product obtained when D-glucose reacts with H_2N-OH. [All India 2015]

Oxime

QUESTION

Define the following term:
Anomers.

[All India 2014, Foreign 2014]

QUESTION

Name two components of starch.

[Delhi 2013C]

Amylose
Amylopectin

QUESTION

Write a reaction which shows that all the carbon atoms in glucose are linked in a straight chain.

[All India 2012]

HI

QUESTION

What is meant by invert sugars?

[Foreign 2012, 2010; Delhi 2010]

QUESTION

Give an example each of reducing and non-reducing sugars.

[Delhi 2010C]

QUESTION

What are monosaccharides?

[All India 2010]

QUESTION

Write such reactions and facts about glucose which cannot be explained by its open chain structure.

[Delhi 2012; Foreign 2011, 2010; All India 2011C, 2011, 2010C]

Direct Neut

QUESTION

Write chemical reactions to show that open structure of D-glucose contains the following.
(i) Straight chain → HI
(ii) Five alcohol groups → acetic anhydride
(iii) Aldehyde as carbonyl group → Br_2 water

[All India 2020]

QUESTION

What happens when D-glucose is treated with the following reagents
(i) Br_2 water ← Gluconic
(ii) HCN ← cyanohydrin
(iii) $(CH_3CO)_2O$ → pentaacetate.

[All India 2019]

QUESTION

Enumerate the reaction of D-glucose which cannot be explained by the open chain structure.

[Delhi 2014C, 2011C]

QUESTION

Amino acids show amphoteric behaviour. Why?

[All India 2015]

QUESTION

What are biocatalysts? Give an example.

[Foreign 2014, Delhi 2012]

QUESTION

Define a peptide linkage.

[All India 2014C]

QUESTION

Write the name of linkage joining two amino acids.

[All India 2013]

QUESTION

What is the biological effect of denaturation of proteins?

[Delhi 2013]

QUESTION

The specific sequence in which amino acids are arranged in a protein is called its [CBSE 2024]

A Primary structure
B Secondary structure
C Tertiary structure
D Quaternary structure

QUESTION

Hydrolysis of sucrose is called [CBSE 2023]

A inversion
B hydration
C esterification
D saponification

QUESTION

Proteins are polymers of [CBSE 2023]

A Nucleic acids
B Amino acids
C Monosaccharides
D Amines

QUESTION

Which of the following sugar is known as dextrose? [CBSE 2022]

A Glucose
B Fructose
C Ribose
D Sucrose

QUESTION

Amino acids which cannot be synthesized in the body and must be obtained through diet are known as

[CBSE 2022]

A. Acidic amino acids
B. Essential amino acids
C. Basic amino acids
D. Non-essential amino acids

QUESTION

Assertion (A): Proteins are polymers of α-amino acids connected by a peptide bond.

[CBSE, 2023]

Reason (R): A tetrapeptide contains 4 amino acids linked by 4 peptide bonds.

C

QUESTION

Assertion (A): D (+) - Glucose is dextrorotatory in nature.

Reason (R): 'D' represents its dextrorotatory nature.

C

Thank you!!